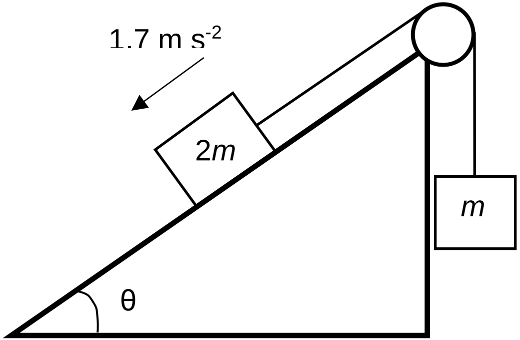
5 The diagram shows two blocks of mass *m* and *2m* connected by a light inextensible cord passing over a light, free-running pulley. At what angle *θ* must the smooth slope be inclined such that the block with mass *2m* accelerates down the slope at 1.7 m s^{-2}?



A 14° **B** 41° **C** 49° **D** 76°

Ans: C

Applying N2L to the 2-masses system,
$2mg \sin \theta - mg = 3m(1.7)$
$2 \sin \theta - 1 = 3(1.7)/9.81$
$\theta = 49°$

6 The diagram shows two spherical masses approaching each other head on at the same